Problem Set 10

Submission:

(Attention: Extended deadline!) Monday, 11/05/2026, until 13:30, HKT (UTC+08:00), to be uploaded on the Canvas course homepage.

Only problems marked with $(*)$ will be graded.

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1. $(*)$ **Multivariate normal distribution** [4 Points]

   Let $X = (X_1, ..., X_d)^\top$ be a random vector with values in $(\mathbb{R}^d, \mathscr{B}(\mathbb{R}^d))$. We say that X has the d-dimensional multivariate normal distribution with $\mu \in \mathbb{R}^d$ and $\Sigma \in \mathbb{R}^{d\times d}$, where Σ is symmetric ($\Sigma^\top = \Sigma$) and positive definite ($u^\top \Sigma u \geq 0$ for all $u \in \mathbb{R}^d$ and $u^\top \Sigma u \Leftrightarrow u = 0$) if it is continuously distributed with the probability density function

   $$f(x) = \frac{1}{(2\pi)^{\frac{d}{2}}\sqrt{\det \Sigma}} \exp\left(-\frac{1}{2}(x-\mu)^\top \Sigma^{-1}(x-\mu)\right), \qquad x = (x_1, ..., x_d)^\top \in \mathbb{R}^d.$$

   We write $X \sim \mathcal{N}_d(\mu, \Sigma)$.

   (a) Show that f is indeed a probability density function.
   Hint: Use from linear algebra that a positive definite symmetric matrix Σ is diagonalizable and has strictly positive eigenvalues. You can also use without proof the following special case of the multidimensional transformation rule:

   $$\int_{\mathbb{R}} g(x)\mathrm{d}x = \int_{\mathbb{R}^d} g(By)\det(B)\mathrm{d}y$$

   for an invertible positive definite matrix B, if the integrals exist.

   (b) Show that $X_1, ..., X_d$ are independent and normally distributed with $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$ if and only if $X \sim \mathcal{N}_d(\mu, \Sigma)$, where $\Sigma = \mathrm{diag}(\sigma_1^2, ..., \sigma_d^2)$.

   (c) Show that for $i, j = 1, ..., d$, one has $\mathbf{E}[X_i] = \mu_i$ and $\mathrm{Cov}[X_i, X_j] = \Sigma_{i,j}$, using the rule $(\star)$ below.

   (d) Prove $(\star)$ for $d = 2$.

   Rule $(\star)$: Let $X \sim \mathcal{N}_d(\mu, \Sigma)$ with $\mu \in \mathbb{R}^d$ and $\Sigma \in \mathbb{R}^{d\times d}$ symmetric and positive definite. Let $p \leq d$, $A \in \mathbb{R}^{p\times d}$ with full rank p, and $b \in \mathbb{R}^p$. Then $AX + b \sim \mathcal{N}_p(A\mu + b, A\Sigma A^\top)$.

2. $(*)$ **Conditional expectation for continuous and discrete random variables** [4 Points]

   Let $(\Omega, \mathscr{F}, \mathbf{P})$ a probability space, and $X, Y : (\Omega, \mathscr{F}) \to (\mathbb{R}, \mathscr{B}(\mathbb{R}))$ random variables.

   (a) Let X, Y be jointly continuous with joint density

   $$f_{X,Y}(x, y) = e^{-y}\mathbb{1}_{\{0\leq x\leq y\}}.$$

   Calculate $\mathbf{E}[X|Y]$ and $\mathbf{E}[Y|X]$.

   (b) Let $r, p \in \mathbb{R}$ and $0 \leq r \leq p \leq 1$ and $1 - 2p + r \geq 0$. The distributions of X and Y are given by

   $$\mathbf{P}[X = 1, Y = 1] = r, \qquad \mathbf{P}[X = 0, Y = 1] = p - r,$$
   $$\mathbf{P}[X = 1, Y = 0] = p - r, \qquad \mathbf{P}[X = 0, Y = 0] = 1 - 2p + r.$$

   Determine the law of Y and the conditional expectation $\mathbf{E}[X|Y]$.

3. **σ-algebras generated by random variables and conditional expectation**

The aim of this problem is to explore the concept of conditional expectation in a more general context than presented in the lecture.

Let $(\Omega, \mathscr{F}, \mathbf{P})$ be a probability space, $Z : (\Omega, \mathscr{F}) \to (\mathbb{R}, \mathscr{B}(\mathbb{R}))$ a random variable with $\mathbf{E}[|Z|] < \infty$, and $\mathscr{G} \subseteq \mathscr{F}$ a sub-σ-algebra. We say that a random variable $U : (\Omega, \mathscr{F}) \to (\mathbb{R}, \mathscr{B}(\mathbb{R}))$ is a *version of the conditional expectation of Z given $\mathscr{G}$* if

(I) U is $\mathscr{G}$-$\mathscr{B}(\mathbb{R})$-measurable, and

(II) For every $A \in \mathscr{G}$, we have
$$\mathbf{E}[Z\mathbb{1}_A] = \mathbf{E}[U\mathbb{1}_A].$$

One writes $U = \mathbf{E}[Z|\mathscr{G}]$. When calculating $\mathbf{E}[Z|\mathscr{G}]$, one can informally view $\mathscr{G}$ as containing all events $A \in \mathscr{F}$ for which the answer "is $\omega \in A$?" is given.

(a) i. Let $\mathscr{G}_1 = \{\varnothing, \Omega\}$. Show that $\mathbf{E}[Z|\mathscr{G}_1] = \mathbf{E}[Z]$. Interpret this result.
    ii. Now suppose that Z is $\mathscr{G}_2$-$\mathscr{B}(\mathbb{R})$-measurable. Show that $\mathbf{E}[Z|\mathscr{G}_2] = Z$. Interpret this result.

To make a connection to the conditional expectation discussed in the lecture, we define for $Y : (\Omega, \mathscr{F}) \to (\mathbb{R}, \mathscr{B}(\mathbb{R}))$ the σ-algebra generated by Y as

$$\sigma(Y) = \{Y^{-1}(B) \,:\, B \in \mathscr{B}(\mathbb{R})\}.$$

(b) Show that $\sigma(Y)$ is a σ-algebra on Ω, and that it is the smallest σ-algebra that makes Y measurable.

From now on, suppose that Z and Y are both discrete random variables.

(c) Show that $\mathbf{E}[Z|\sigma(Y)]$ is given by $\mathbf{E}[Z|Y]$ as defined in the lecture.

(d) Let $(\Omega, \mathscr{F}, \mathbf{P}) = (\{1, ..., 6\}, \mathscr{P}(\{1, ..., 6\}), \mathscr{U}(\{1, ..., 6\}))$ and

$$Z(\omega) = \omega,$$
$$Y(\omega) = \mathbb{1}_{\{1,3,5\}}(\omega)$$

   i. Write down $\sigma(Y)$ explicitly and explain its interpretation.
   ii. Calculate $\mathbf{E}[Z|Y]$.
   iii. Now let $X(\omega) = \mathbb{1}_{\{1,2\}}(\omega)$. Calculate $\mathbf{E}[Z|X]$ and $\mathbf{E}[Z|\sigma(X, Y)]$, where $\sigma(X, Y) = \sigma(\sigma(X) \cup \sigma(Y))$. Compare the values and explain the difference.

4. **Random powers of random variables**

Let X and Y be two independent random variables following Poisson distributions with parameters α (> 0) and β (> 0), respectively. Consider $Z = X^Y$.

(a) Calculate $\mathbf{E}[Z]$.
Hint: It is useful to condition on the values of one of the random variables.

(b) Note that $\mathbf{E}[Z]$ can be viewed a function of α and β. Does it increase as α and β increase?